

# What AcuPebble, MAM, Dual A, and PSG actually measure

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**Ox lock:** AcuPebble **always** uses finger ox (AHI + ODI). MAM, ANR, and Dual A **do not**. Dual A SpO<sub>2</sub> is temple PPG. Oralable does not replace AcuPebble AHI.

## Why this note

You already run **AcuPebble** for home OSA. Oralable (**MAM**) sits on the **temple**. ANR is **EMG** on the same temporalis belly. Dual A is MAM + ANR on that site. These are different physics. Mixing them in methods copy makes AHI look like temple SpO<sub>2</sub>, or EMG look like an IR trough.

### Three oxygen numbers (do not merge):

| Name                           | What it counts                                    | Needs scored apneas/hypopneas?                    |
|--------------------------------|---------------------------------------------------|---------------------------------------------------|
| AHI                            | Event <b>count</b> per hour of sleep              | Yes                                               |
| HB (Azarbarzin hypoxic burden) | SpO <sub>2</sub> <b>area tied to those events</b> | <b>Yes</b>                                        |
| SASHB                          | Oralable area when temple SpO <sub>2</sub> < 90%  | <b>No</b> — temple PPG only, <b>not</b> finger ox |

Your AcuPebble headline is **AHI / ODI**. That is not Azarbarzin HB unless your export has event-tied SpO<sub>2</sub> area. We do not claim that.

## Relevance to you

| Your work              | What the tables say                                                                                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AcuPebble (today)      | Home AHI + ODI from neck + <b>finger</b> SpO <sub>2</sub> . Keep this as the apnea reference. Oralable does not replace it.                                                                     |
| Arm P / MAD nights     | MAM can sit beside AcuPebble for temple vitals and later jaw-load. Temple SpO <sub>2</sub> ≠ finger ox. SASHB ≠ AHI ≠ HB.                                                                       |
| Dual A (research)      | Same-site EMG burst → IR-DC trough (~1–5 s lag). One engineering pack: IR↔EMG F1 ≈ 0.61 vs EMG; F1 vs Protocol A labels = 0; ~6 min, not a night.                                               |
| FEP / hypothesis 3.5.4 | Needs <b>event timestamps</b> plus a bout clock. Nightly AHI is not enough. Closest home path: AcuPebble events + Dual A. <b>Not built</b> . Cite FEP for Paper B / Arm P, not Paper A methods. |
| PSG-AV                 | Gold AHI, RMMA, and HB if scored. Lab, not the home stack we ship.                                                                                                                              |

Paper A is feasibility / Dual A precursor. It does **not** test FEP endotypes.

## Table 2 — what each stack gives you now

| Stack                 | You get now                                                                                                    | You still lack                                                                                                 |
|-----------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| MAM (Oralable temple) | IR trough, % drop, SASHB, tonic/phasic/rescue/recovery <b>labels</b> , Core ML, <b>temple</b> SpO <sub>2</sub> | Finger ox; AHI; HB; electrical onset; FEP latency                                                              |
| ANR                   | EMG onset + amplitude bouts                                                                                    | Finger ox; AHI; HB; IR; Core ML; rescue-as-airway                                                              |
| Dual A                | EMG→IR trough pair + <b>temple</b> SASHB shown beside EMG                                                      | Finger ox; AHI; HB; Core ML vs EMG; overnight tonic/phasic F1; FEP latency                                     |
| AcuPebble             | AHI + ODI / <b>finger</b> SpO <sub>2</sub> ( <b>always</b> )                                                   | HB unless export is event-tied; all jaw rows                                                                   |
| AcuPebble + Dual A    | AHI <b>plus</b> EMG→IR. Finger SpO <sub>2</sub> is <b>AcuPebble only</b> ; Dual A still temple + EMG           | Coupling only if event timestamps align — <b>not built</b> . Dual A SpO <sub>2</sub> is not the finger channel |
| PSG-AV                | AHI <b>and</b> HB (if scored) <b>and</b> RMMA                                                                  | Oralable IR-DC / Core ML / SASHB unless MAM is worn                                                            |

Table 1 — constructs (what matches what)

**Now** = what exists today. **MAM if verified** = the name we may use only after Dual A (or PSG) F1, or an event clock. **Still not** = even after a pass.

| #  | Construct        | Now                                                         | PSG-AV                              | AcuPebble                                                     | ANR                             | Dual A now                                                    | MAM if verified                                                       | Still not                                             |
|----|------------------|-------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------|---------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------|
| 1  | AHI              | Impossible from MAM/ANR                                     | Gold                                | Home AHI                                                      | No                              | No                                                            | None                                                                  | Temple PPG/ACC/IR cannot become AHI                   |
| 2  | HB (Azarbarzin)  | Impossible (no finger ox, no scored events)                 | Yes if event-tied finger/PSG area   | AHI/ODI ≠ HB unless export is event-tied — <b>not claimed</b> | No                              | No                                                            | None as HB                                                            | SASHB; Dual A pairing; AHI; ODI                       |
| 3  | SASHB            | Temple PPG SpO <sub>2</sub> <90 area — <b>not finger ox</b> | Not this formula                    | Not AcuPebble's formula                                       | No (no SpO <sub>2</sub> )       | Temple SASHB shown beside EMG. Not AHI, not HB, not finger ox | Stay “temple SpO <sub>2</sub> <90 area”                               | AHI; HB; ODI; finger SpO <sub>2</sub>                 |
| 4  | LP IR-DC trough  | Scored Dual A                                               | No optical OMG                      | No                                                            | No                              | <b>Primary pair:</b> EMG first, IR later                      | Occlusion onset / trough (~1–5 s lag)                                 | EMG onset; μV                                         |
| 5  | EMG burst onset  | ANR / PSG                                                   | <b>Yes</b> — RMMA clock             | No                                                            | <b>Yes</b>                      | Clock for row 4                                               | MAM cannot get electrical onset                                       | Millisecond lockstep                                  |
| 6  | IR-DC % drop     | Gate ≥8% of rest                                            | No                                  | No                                                            | No                              | EMG ≥70 <b>and</b> IR ≥8%                                     | Occlusion depth (% of rest)                                           | EMG amplitude; %MVC                                   |
| 7  | EMG amplitude    | ANR 0–1023                                                  | μV / often ≥10% MVC                 | No                                                            | <b>Yes</b>                      | Contact proof, not MVC                                        | —                                                                     | Depth ≠ μV                                            |
| 8  | Overnight tonic  | MAM label                                                   | Tonic RMMA (different def)          | No                                                            | Sustained EMG, not scored vs IR | Not F1'd                                                      | Tonic occlusion minutes                                               | Tonic RMMA until F1                                   |
| 9  | Overnight phasic | MAM label                                                   | Phasic RMMA                         | No                                                            | Phasic EMG                      | <b>Not F1'd</b> — ACC ≠ EMG                                   | High-motion bouts; this row may fail                                  | Phasic RMMA; grinding EMG                             |
| 10 | Core ML MAM      | Product classes; 10% IR-DC <b>gates</b> only                | No                                  | No                                                            | No                              | Never run vs ANR                                              | 1 s optical–motion class if F1 vs EMG holds                           | EMG class; apnea class                                |
| 11 | Rescue           | MAM state. EMG ∩ <b>temple</b> desat. <b>No finger ox</b>   | Events + RMMA (finger ox)           | AHI events + <b>finger</b> desat if exported                  | No                              | Pairing ≠ rescue = EMG                                        | Load + temple-desat bout if timestamps match AcuPebble/PSG event ends | AHI; finger-ox desat; FEP airway RMMA until scored    |
| 12 | Recovery         | IR-DC + ACC settle                                          | Could define EMG + autonomic return | No                                                            | Unused for this                 | Not FEP latency                                               | Optical–motion return time                                            | <b>Homeostatic latency</b> (your paper: EMG + HR/HRV) |
| 13 | TFI              | Optical load index                                          | No                                  | No                                                            | No                              | No                                                            | Night occlusion burden if vs EMG episode index                        | MVC%; AASM SB index                                   |

**Engineering Dual A pack (locked, not a night):** session 20260812\_085110, ~6 min. Median EMG→IR lag ≈ **4.9 s**. IR↔EMG F1 ≈ **0.61**. F1 vs Protocol A labels = **0**.

Do / do not (for methods)

| Do                                                            | Do not                                                                               |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Keep AcuPebble for AHI / ODI (finger ox)                      | Call Oralable or Dual A an HSAT or finger-ox channel                                 |
| Call Dual A primary pair EMG → IR trough                      | Call the IR trough EMG; call ACC phasic RMMA                                         |
| Call tonic / phasic / rescue / Core ML <b>labels</b> until F1 | Treat Core ML as scored vs ANR                                                       |
| Call recovery optical–motion settle                           | Call it homeostatic latency                                                          |
| Call AcuPebble AHI/ODI                                        | Call AHI or ODI Azarbarzin HB unless the export has event-tied SpO <sub>2</sub> area |

Device-inferred wellness / research signals — not a medical diagnosis. Paper A does not test FEP. Canonical lock: MEASUREMENT\_CONSTRUCT\_MAP (data room). Do not paste Table 1 into Paper A methods — point here.